Code No: 155AA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, August - 2022
ADVANCED COMPUTER ARCHITECTURE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Describe various shared-memory multiprocessor models in detail.
b) Explain the layered development of parallel computers. [8+7]

2.a) Illustrate with an example detection of parallelism in a program using Bernstein's condition.
b) Distinguish between control flow and data flow architectures. [8+7]

3.a) What are various scalability metrics? Explain.
b) What do you mean by efficiency, utilization and quality of parallelism? Explain. [8+7]

4.a) Distinguish between RISC and CISC architectures.
b) What do you mean by inclusion, coherence and locality? Explain. [8+7]

5.a) Explain backplane bus specification.
b) Explain interleaved memory organization in detail. [8+7]

6.a) Explain various branch handling techniques in instruction pipeline.
b) Describe sequential consistency model. [8+7]

7.a) Explain snoopy bus protocol.
b) What are compound vector operations? Explain with an example. [8+7]

8.a) What do you mean by shared virtual memory? Explain.
b) What are various issues in multithreading? Explain. [8+7]

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